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AI & Blockchain Integration

Learning Objectives

- Understand blockchain architecture and how it connects to AI
- Compare consensus mechanisms like PoW and PoS
- Identify real-world uses of AI and blockchain
- Understand challenges and future trends

What is Blockchain?

Blockchain is a decentralized digital ledger used to record transactions securely.

Key Features

- Distributed Ledger Technology (DLT) – data is shared across many computers
- No central authority controlling the network
- Nodes validate transactions

Block Structure

Each block contains:

- Header – timestamp and transaction information
- Cryptographic links – hashes connecting blocks securely

Blockchain History

Major developments include:

- Early distributed systems before Bitcoin
- Bitcoin introduced in 2008
- Ethereum launched in 2015 with smart contracts

Blockchain Architecture Types

Centralized

- Controlled by one authority
- More vulnerable to attacks

Distributed

- Data shared across multiple computers

Decentralized

- No central authority
- More transparent and secure

Consensus Mechanisms

Consensus mechanisms are used to validate transactions and maintain trust.

Proof of Work (PoW)

- Miners solve complex puzzles
- First miner to solve it receives a reward
- Requires high computational power

Proof of Stake (PoS)

- Validators selected based on their holdings
- Uses much less energy than PoW

Other Mechanisms

- DPoS – Delegated Proof of Stake
- PBFT – Practical Byzantine Fault Tolerance

Blockchain Security

Important security features include:

- **Cryptography** to secure transactions
- **Immutability** meaning data cannot be changed once recorded
- **Decentralization** which spreads data across many nodes

How Blockchain Improves AI

Blockchain helps AI systems by providing:

- **Data integrity** so AI models use reliable data
- **Audit trails** to track AI decisions
- **Decentralized computing** for distributed processing

Real-World Applications

Healthcare

- Secure patient data sharing
- AI diagnostics

Finance

- Fraud detection
- Smart contracts

Supply Chain

- Tracking goods
- AI demand prediction

Government

- Transparent voting systems

Education

- Credential verification

Case Study: IPwe Patent Registry

Problem:

- Patent management was slow and inefficient

Solution:

- **Blockchain** created a transparent and secure ledger
- **AI** analyzed patent value and trends

Benefits:

- Improved visibility
- Fraud prevention
- Faster patent transactions

Case Study: Walmart

Blockchain helps track food origin and safety.

Benefits:

- Faster contamination detection
- Better supply chain transparency

AI helps with:

- Inventory optimization
- Demand forecasting

Challenges

- **Scalability** issues with large transaction volumes
- **Interoperability** between different systems
- **Ethical concerns** about transparency and privacy

Future Trends

- Decentralized AI marketplaces
- Federated learning with blockchain
- Edge computing combined with blockchain security

Ethical Considerations

- Addressing AI bias
- Protecting data privacy
- Developing global governance standards

Key Takeaways

- Blockchain improves security and transparency
- AI benefits from trusted data and traceability
- Together they can transform industries and improve trust in technology